

Uberny ML & DL, NLP course

①

~~see~~ ~~lec~~ - 35
Adaboost ML Algorithm

lec - 1

Introduction to Adaboost
ML Algorithm

lec - 2

Creating Decision Tree
Stump

lec - 3

Performance of Decision
Tree Stump

Decision Tree

Ensemble techniques

Bagging

① Random Forest
(Classifier)

② Random Forest
Regressor

Boosting

① Adaboost

② Gradient
Boosting

③ XgBoost

Decision Tree:-

Overfitting:

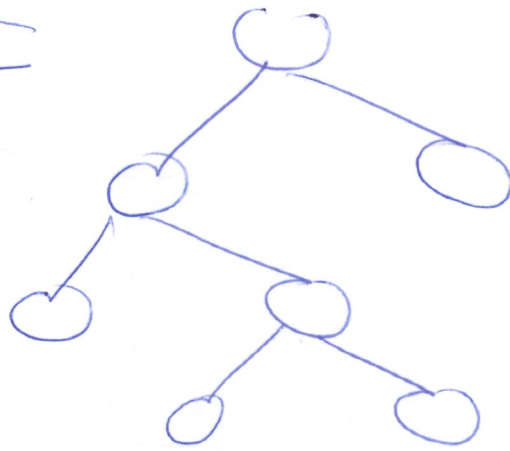
Training
Acc ↑

Test Acc ↓

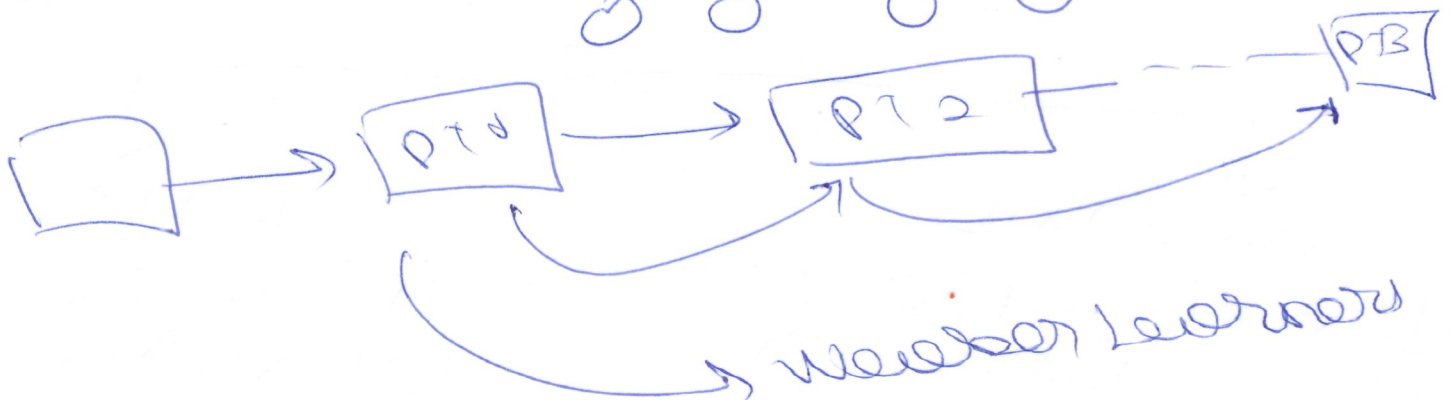
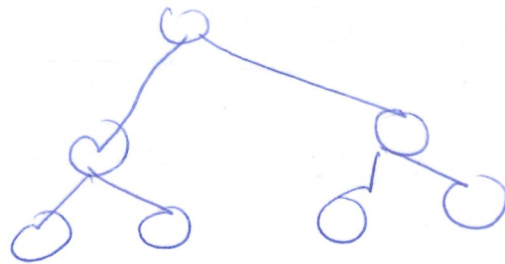
Learned things wrong

⇒ Random Forest

→ our trees select samples
as it is



Boosting:-



Weak learners :-

(3)

↳ Having learnt
much from training
dataset

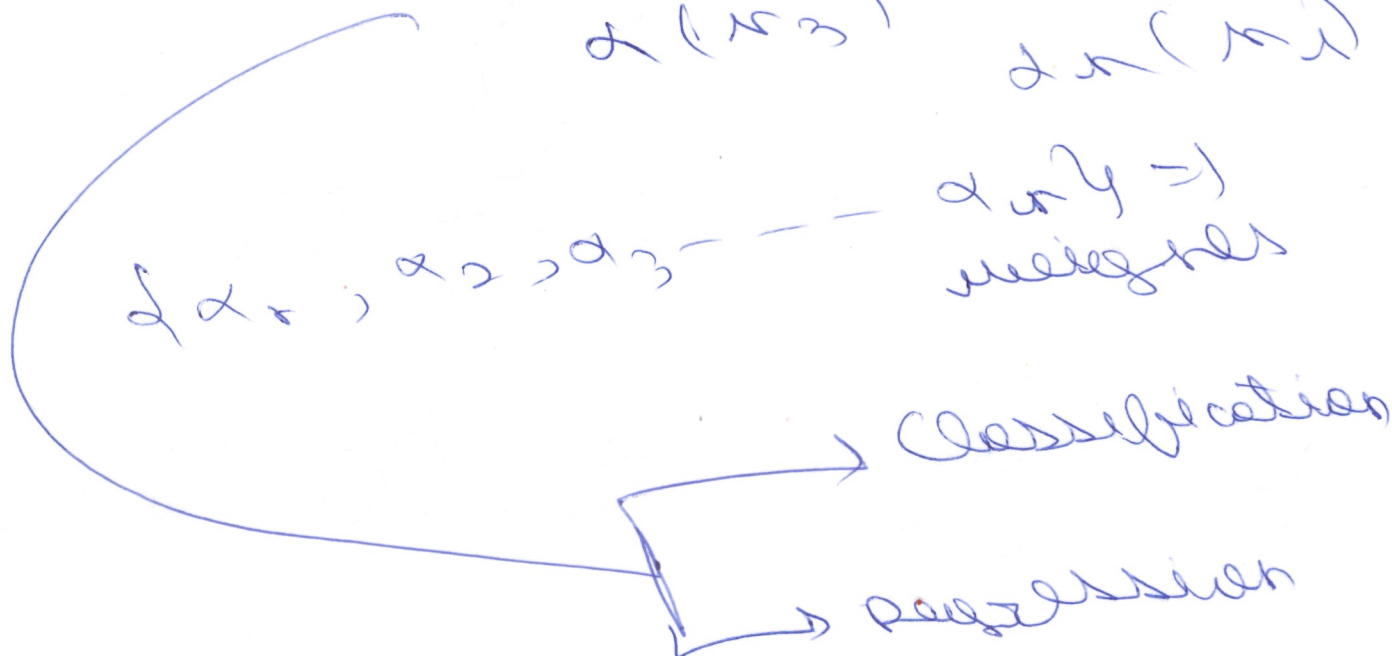
Random Forest :- majority
voting classifier

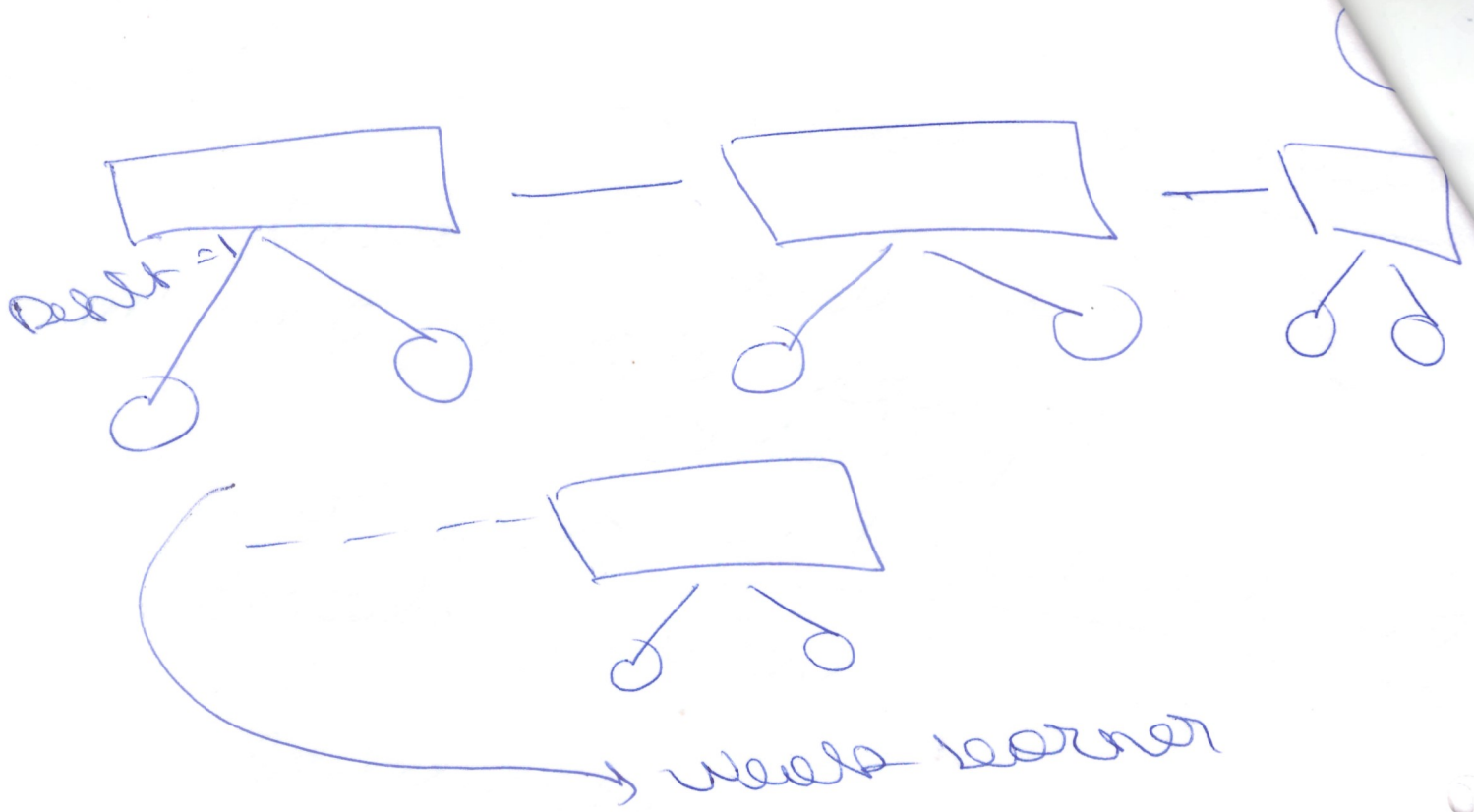
Ada Boost :- Assignment
weights to the
weak learner

$$f = \alpha_1(m_1) + \alpha_2(m_2) +$$

$$\alpha_3(m_3) + \dots + \alpha_n(m_n)$$

$\{\alpha_1, \alpha_2, \alpha_3, \dots, \alpha_n\}$ — $\alpha_i \neq 0$
weights





underfitting

train accuracy ↓ 40%
test accuracy ↑ 25%

High Bias
Low Variance

Low Bias
High Variance

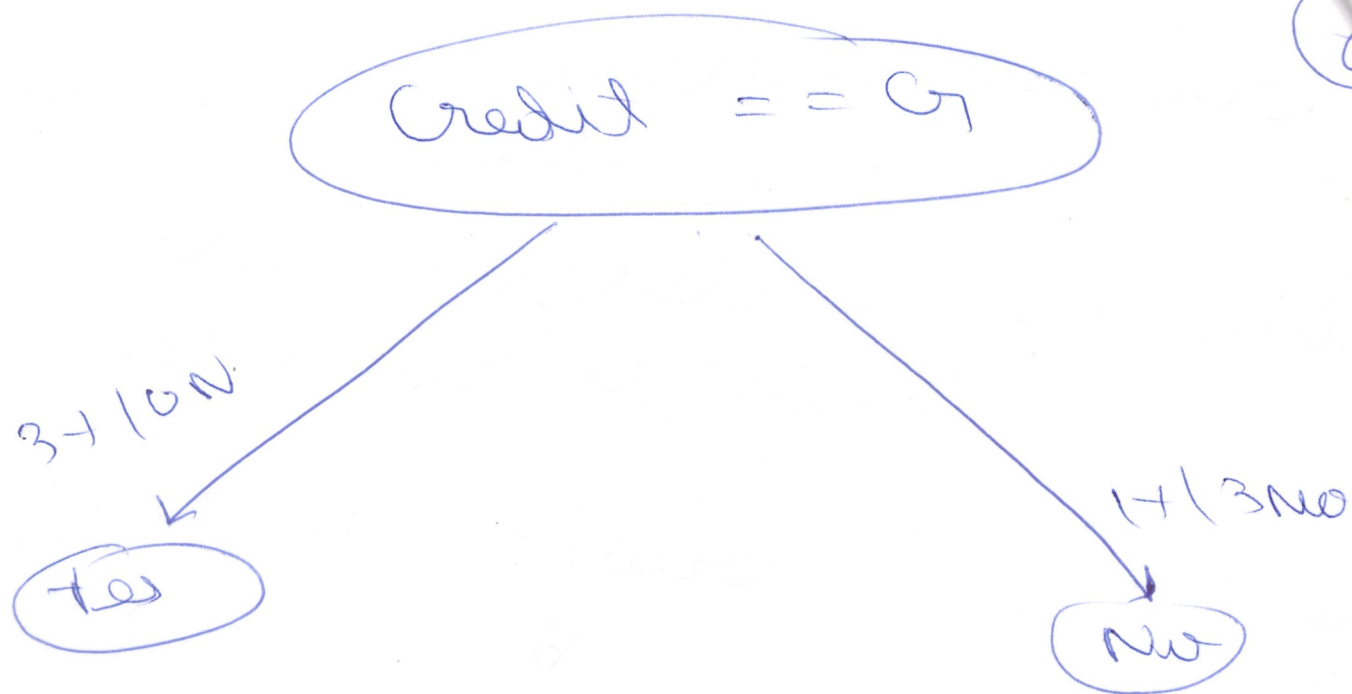
We combine all decision
tree terms along with weights

→ Creating Decision Tree (5)

Adaboost Classifier using
Indepth Intuition.

Salary	Credit	Approval
$\leq 50K$	B	No
$\leq 50K$	G	Yes
$\leq 50K$	G	Yes
$\leq 50K$	B	No
$> 50K$	B	Yes
$> 50K$	G	Yes
$> 50K$	N	No
$\leq 50K$		



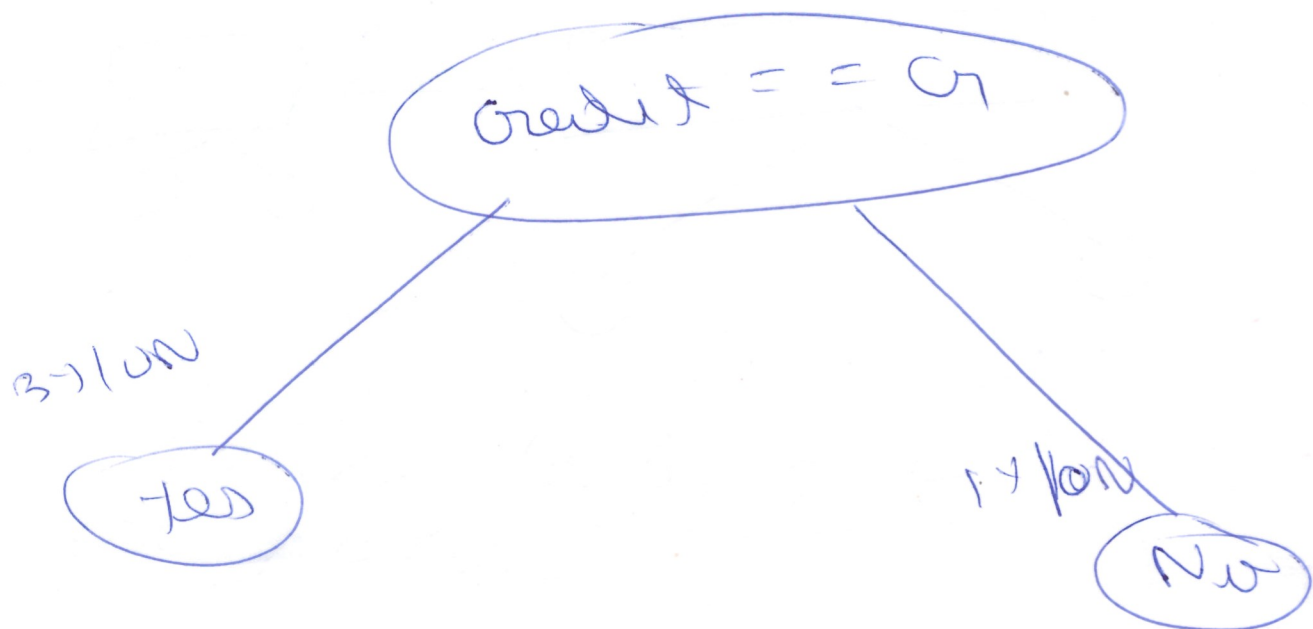


Entropy or Gini Impurity

$$H(S) = -p + \log_2 p - p - \log_2 p$$

we select the best split
using Entropy or Gini
Impurity.

Selector	Credit	Approval	Sample weight
$L = SOK$	B	No	1/7
$L = SOK$	G	Yes	1/7
$L = SOK$	G	Yes	1/7
$L = SOK$	G	Yes	1/7
$> SOK$	B	No	1/7
$> SOK$	G	Yes	1/7
$> SOK$	22	Yes	1/7
$L = SOK$	22	No	1/7



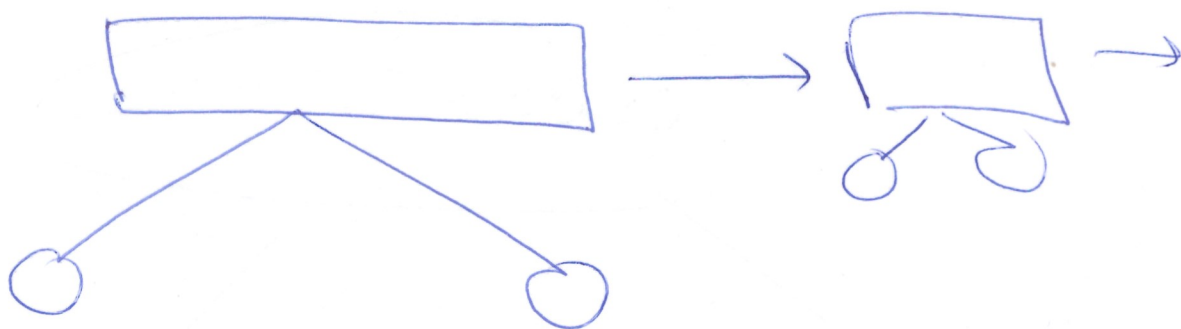
=> sum of all error

① Total Error = $\frac{1}{2}$

② performance of stump = $\frac{1}{2} \ln \left[\frac{1 - TE}{TE} \right]$
 $= \frac{1}{2} \ln \left[\frac{1 - \frac{1}{2}}{\frac{1}{2}} \right]$

performance of stump ≈ 0.896 $= \frac{1}{2} \ln(6)$
 $= 0.896$

$f = \alpha_1(m_1) + \alpha_2(m_2) + \dots + \alpha_n(m_n)$
 $\alpha_1 = 0.896$



predict same
 wrong answer